

Discord Redesign Phase 5

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Abstract— Over the past eight years since its initial release date, Discord has become a prevalent communication social platform in which countless people use on a daily basis. The main functionalities of Discord include instant messaging, voice calls, and video calls for individuals and groups. The main task that I will be discussing involves Discord's group voice channels for group users.

1 EMPIRICAL EVALUATION

For the empirical evaluation, the verbal prototype was used in the process. The evaluation was done based on the measurement of time that was taken during the activity involving the participants. The participants were given two different images of Discord's interface, one of the current design and one with the design of the prototype. The images were given to the participants in a random order and were tasked to identify where to click on the interface to be able to join a group voice channel. The time the participants took to complete this task was taken for analysis.

1.1 Testing Process

After the sessions were completed, there were no noticeable issues that had arisen during the process. There were a total of five participants in total, and each participant had no previous experience with Discord. Each session with the participants went fairly smoothly and times of all the participants was successfully recorded. The only difference between each session of the participants were the physical location that the session took place and the order in which the images were shown. The images were purposely randomly selected so this did not cause any problems to arise. However, since the physical location of each participant could not be ensured to be isolated, there could have been in person environmental factors that slightly altered a participant's time. Next time, a better implementation of the sessions would be to have an isolated area where all the participants would be ensured the exact same settings for each session would prevent any environmental factors to affect the data.

1.2 Results

The raw data of the testing process can be seen in the appendix. Before looking at the results of the t test, simply looking at the times of the participants, there is a significant increase in time that a participant requires to complete the task of locating how to join Discord's voice channel. Even with the images being randomly given, the time required by the individuals tends to greatly increase when working with the image of Discord's current interface. The results of the t test shows that there is a confidence interval from 11.07 to 26.53. The p-value of the t test is 0.0005. According to this data, there is a considerable difference in the data resulting in the conclusion that the difference is statistically significant. Based on the test, I can confidently reject the null hypothesis that the time a new user takes to complete the task of finding the button to join Discord's voice channel is the same for both interface designs.

1.3 Analyze

With the data and the t test, I reject the null hypothesis which completely supports my original theory that this new design will allow new users to better learn how to use Discord's group voice channel. I expected that the data from the testing would support my alternative hypothesis; however, I did not foresee such a wide gap in time new users would take to complete the task. The results clearly supported the new interface design; however, the results also showed that since the images of the design were shown to the participants one after the other, the participants would tend to have a lower time on the second image. This experimental oversight could have caused the times for the second images to be less than what it should have been. However, since the order the images were given were random, this issue was somewhat mitigated.

1.4 Changes

Based on this analysis, I believe this design was a definitive improvement to Discord's current interface. Although this was a minor change, the results of new users taking much less time to learn how to use the interface was clear. This specific design will definitely be added into the next round of prototyping along with the characteristic of clearly defining major tasks within the interface, such as clearly labeling the "Leave" call button or any other important buttons.

2 PREDICTIVE EVALUATION

The predictive evaluation was done utilizing the textual prototype where the design had a few minor changes to the original interface. The prototype encompassed the addition of a “Join” and “Leave” button for the user. The cognitive walkthrough fully encompasses every aspect of how the user is able to join and leave Discord’s group voice channel successfully.

2.1 Cognitive Walkthrough

The cognitive walkthrough can be broken down into two sections for the entire process. The cognitive walkthrough begins with the user already logged into Discord and is currently on Discord’s main page so the walkthrough will not involve any aspects of logging into Discord itself.

2.1.1 First Process

The first part of the process requires the user to control the computer mouse to be able to click on the correct button. Since the user should be familiar with how to use a computer if the individual is using Discord, the cognitive load on the user for this operator should be minimal. Looking at the prototype design interface, the user should be the standard original interface of Discord. The only change would be the placement of a “Join” button next to channel names of all the channels in the server. To complete the first process of the cognitive walkthrough, the user would simply move the mouse over to this button and click on the button. The user’s task of joining a group voice channel would be complete. The user will be able to understand and know how to complete this first task through the design principle of simplicity and mapping. The button’s mapping is key to relieving the user’s cognitive load. With the location of the button right next to the name of the group channel names, the user will be able to easily associate each button with its individual channels. In addition, the simplicity of the button also assists the user in knowing what clicking the button entails. By clearly labeling the button to say “Join” the user will instantly know what the button allows the user to join, which is the voice channel that is right next to the button.

2.1.2 Second Process

The second part of the process involves the task of leaving the current voice channel. The design of the prototype incorporates a “Leave” button for the users

to utilize. This button will replace the previous “Join” button that the user used to enter the voice channel if the user is currently inside the specified channel. In addition, the color of the “Leave” button will be red to further indicate to the user its use. For this process, the user will continue to use the mouse to click on the “Leave” button once he or she wants to complete the voice channel call. The user’s cognitive load is lessened in this task through the design principles of simplicity, mapping, and discoverability. The design of the button upholds the simplicity through its use of directly telling the user what it will do. This allows the users to not have increased cognitive load having to work out what the button will do. Similarly, the mapping of the button is crucial in helping the user as well. Since the location of the button maps the button to each individual group voice channel, the individual will have a much easier time understanding their association. Discoverability is the last design principle that this button incorporates. Since the color of the button will be changed to red, this draws the user’s attention to the sudden change. The gray “Join” button changing to a red “Leave” button will immediately draw the user’s attention and allow the user to identify where he or she would need to click to leave the voice channel properly. All of these aspects allow the user’s cognitive load to be lessened so that he or she can focus solely on the task itself.

3 EVALUATION SUMMARY

3.1 Next Iteration

The next iteration of the design life cycle will include more drastic changes in the original Discord interface design along with the current changes. Based on the results of the evaluation, the addition of the “Join” and “Leave” buttons from design 1 and 2 will be able to drastically improve the interface’s learnability for new users. However, the next step in the life cycle will also try to improve the design’s human error tolerance as well. To further this goal, additional needfinding will be crucial. The use of a loud and post event protocol needfinding exercises will be helpful in determining if the learnability of the current interface is sufficient and whether or not the error tolerance needs to improve as well.

3.2 Evaluation

After the results of the evaluation, it has been made clear that the design changes to the interface can be improved further. Although the changes showed clear

improvements, the design was only minorly changed from the original. This showed more revisions to the interface along with the current changes can be possible with little repercussion. Possible major design changes could be the entire redesign of the Discord group voice channel grid box that lists all the voice channels. By adjusting the design, this can reduce the repetition of the buttons occurring multiple times in a list while maintaining the current prototype's learnability.

3.3 Brainstorm

The current evaluations are overwhelmingly positive for the new designs. However, I believe that changing the prototypes further can help them resolve much more and further improve learnability. I would make changes to the overall layout of Discord's voice channel grid design to make buttons much more explicit with better mapping. This could greatly improve a new user's experience; however, this requires me to change the current design at this level of fidelity if I do not want to risk a possible bad design having negative repercussions.

3.4 Plan

With these changes in mind, the best way to evaluate these changes would be to have another empirical evaluation. The empirical evaluation for the current prototypes was able to best demonstrate whether or not the new interface design was an improvement. With such a radical change with the new design and prototype, I believe the best way to properly compare the newest design to the current design is to use empirical evaluation.

4 APPENDICES

4.1 Empirical Evaluation Raw Data

- Participant 1
 - Image Order:
 - Discord Current Interface (Image D)
 - Prototype (Image P)
 - Time:
 - Image D: 27 seconds
 - Image P: 3 seconds
- Participant 2

- Image Order:
 - Discord Current Interface (Image D)
 - Prototype (Image P)
 - Time:
 - Image D: 36 seconds
 - Image P: 6 seconds
- Participant 3
 - Image Order:
 - Prototype (Image P)
 - Discord Current Interface (Image D)
 - Time:
 - Image D: 22 seconds
 - Image P: 10 seconds
- Participant 4
 - Image Order:
 - Discord Current Interface (Image D)
 - Prototype (Image P)
 - Time:
 - Image D: 19 seconds
 - Image P: 4 seconds
- Participant 5
 - Image Order:
 - Prototype (Image P)
 - Discord Current Interface (Image D)
 - Time:
 - Image D: 20 seconds
 - Image P: 7 seconds